Applied AI & Machine Learning
CS-333

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PNEC, NUST

Lecture 7

Spring 2026

AI-Based Image Analysis in Computer Vision

Learning Outcomes

By the end of this lecture, students will be able to:

- Explain the fundamentals of image analysis in computer vision systems.
- Explain image classification methods for assigning labels to visual data.
- Describe object detection techniques used to identify and localize objects in images.
- Understand the principles of image segmentation for region-based analysis.
- Understand the role of pattern recognition in computer vision.

Topics

1. Image Representation
2. Image preprocessing
3. Feature Extraction
4. Classification
5. Object Detection
6. Segmentation

What is Computer Vision?

- It helps machines understand and interpret visual information like humans.
- Applications span industries like healthcare, automotive and entertainment.
- Revolutionizing object recognition, face detection and more.

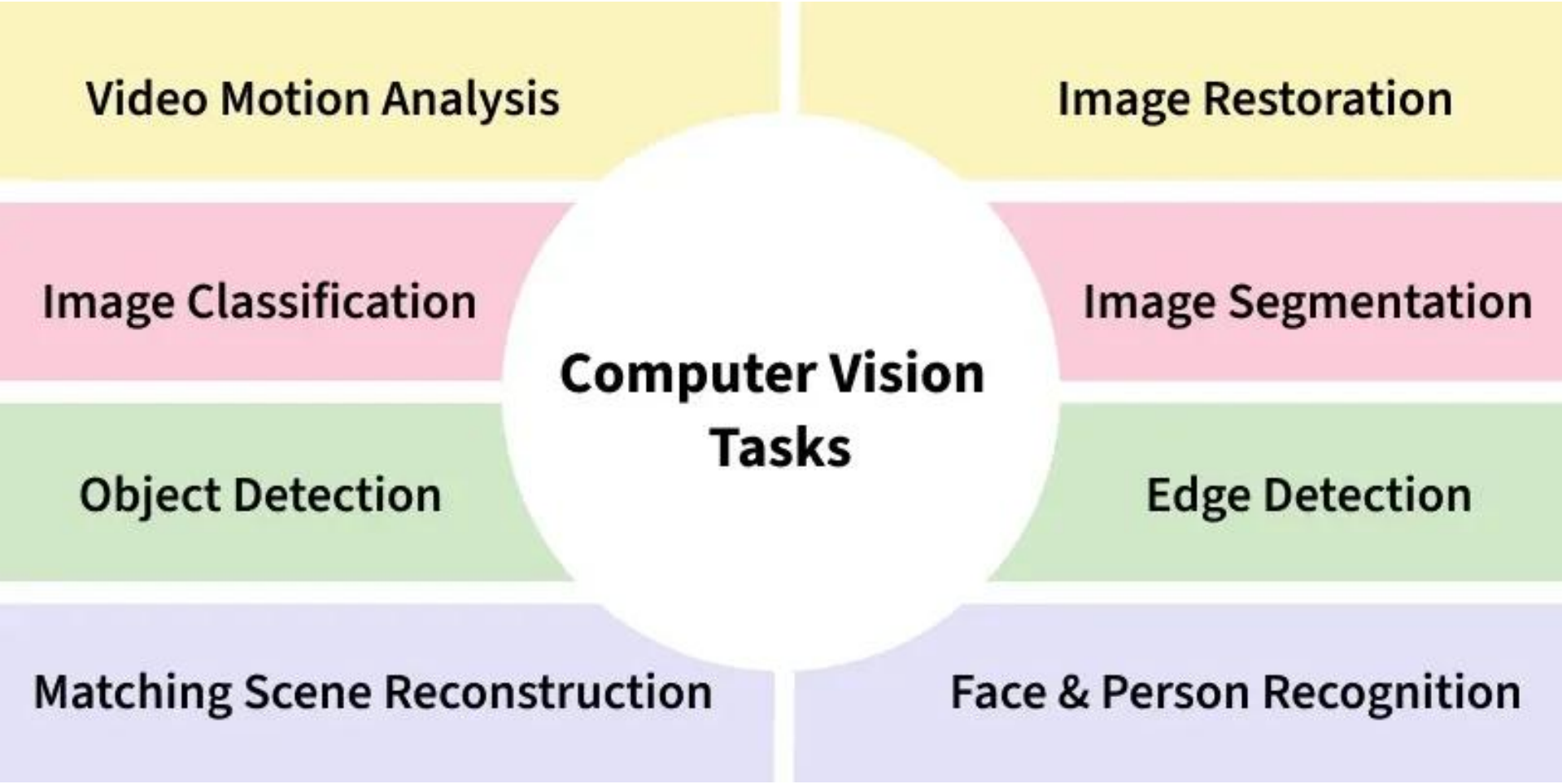

Video Motion Analysis
Image Restoration
Image Classification
Computer Vision Tasks
Image Segmentation
Object Detection
Edge Detection
Matching Scene Reconstruction
Face & Person Recognition

Image Analysis is the process of extracting meaningful information from images using computational techniques.

Key goals:

- Identify important structures
- Extract useful features
- Interpret visual information

Applications:

- Medical imaging
- Autonomous vehicles
- Industrial inspection
- Facial recognition

Image Analysis in Computer Vision Pipeline

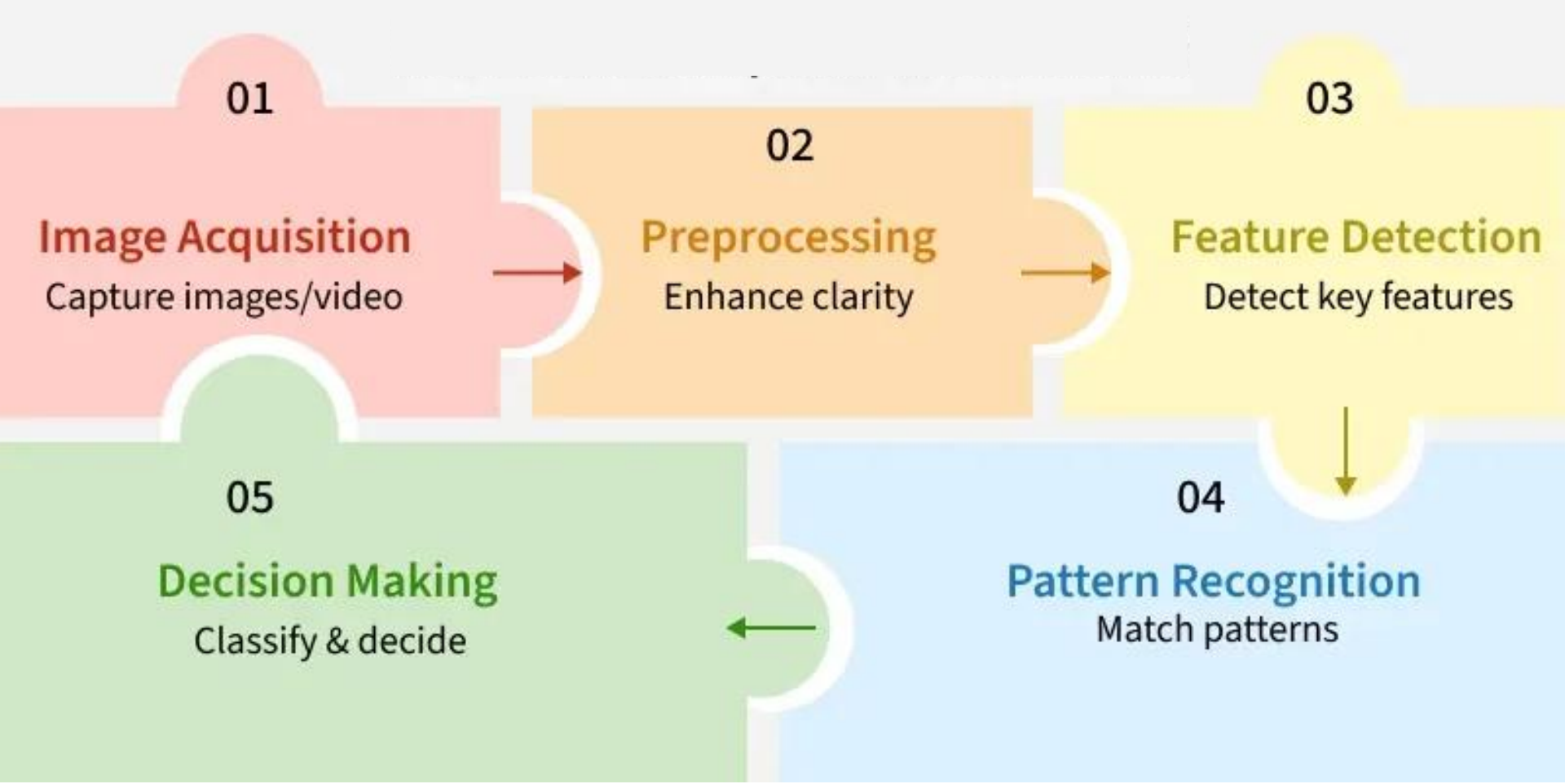


Image Representation

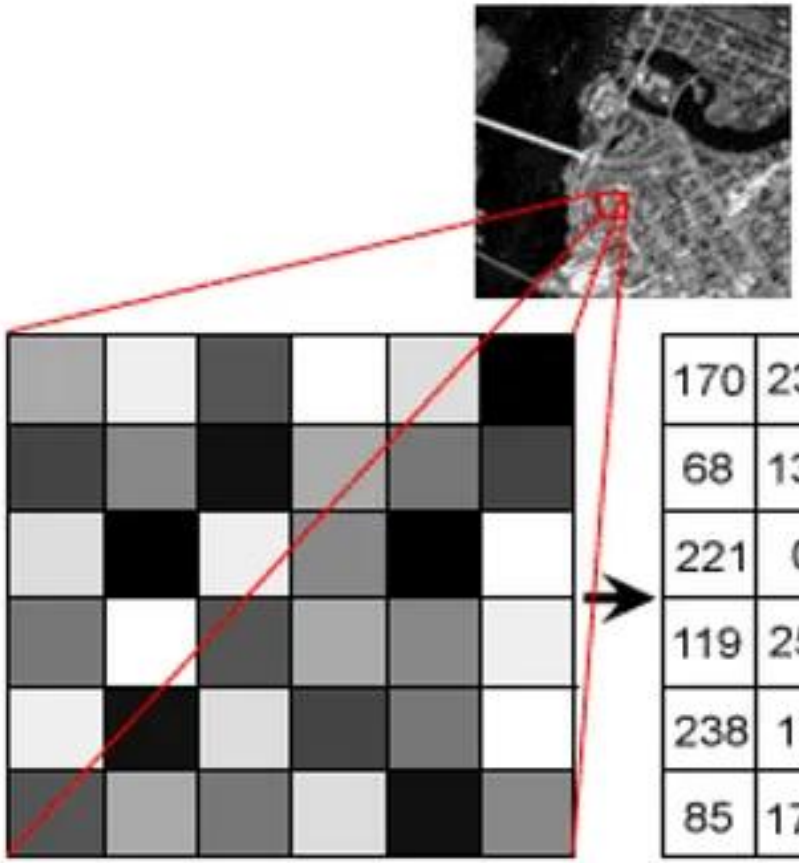
Pixels: The Atomic Units of Digital Images

- It is the smallest units of an image that can be displayed and processed on a digital display.
- Each pixel represents a dot or a square of color in an image.
- In color images, pixels are typically defined by three components: red, green, and blue (RGB), each varying in intensity from 0 to 255.
- This RGB model allows for the creation of over 16 million distinct colors by mixing these three primary colors in different proportions.

Image size is determined by two key factors: **resolution and aspect ratio**

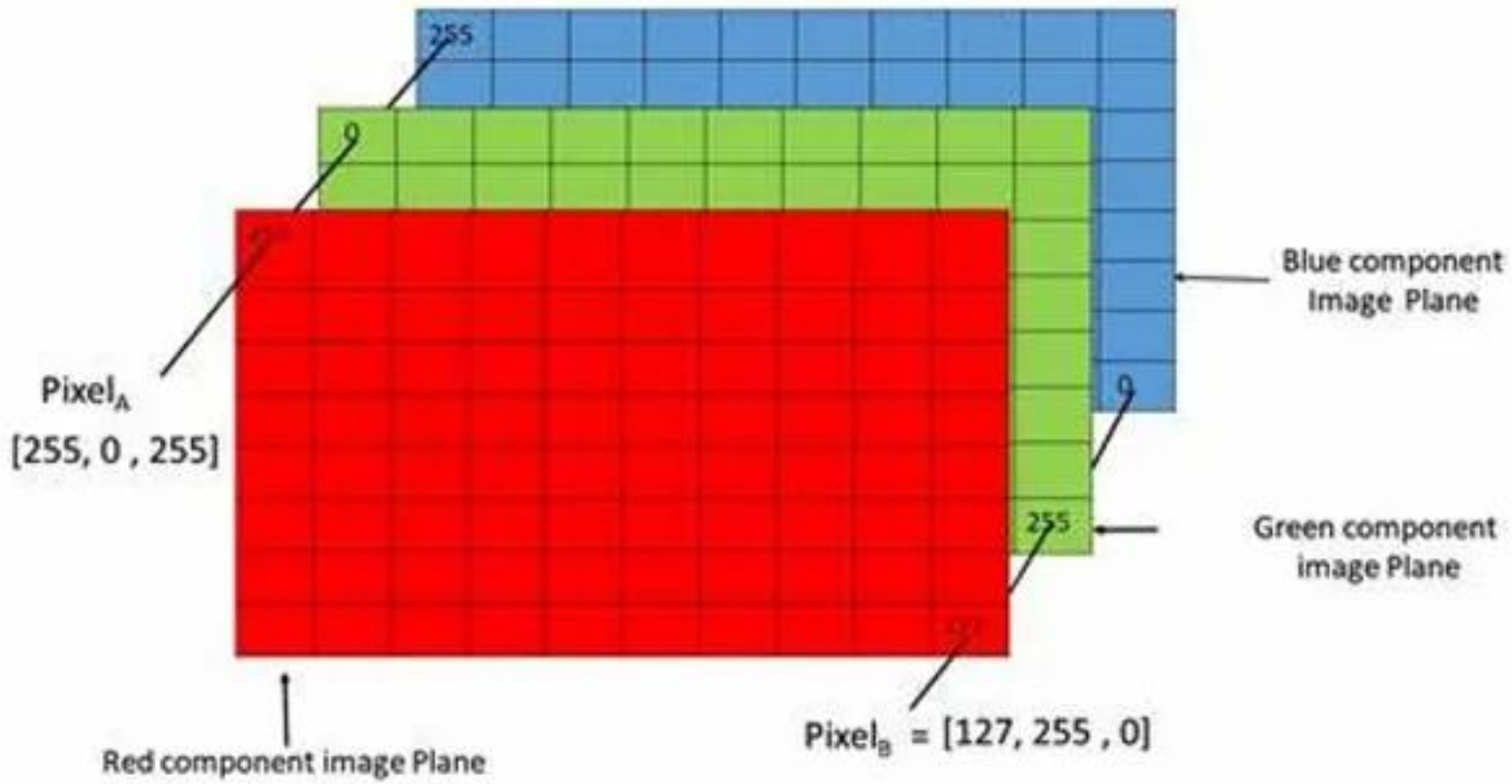
- **Resolution** refers to the number of pixels in an image, usually denoted as width × height.
- **Aspect ratio** describes the proportional relationship between an image's width and height, dictating its shape and framing.

Image Representation



170	238	85	255	221	0
68	136	17	170	119	68
221	0	238	136	0	255
119	255	85	170	136	238
238	17	221	68	119	255
85	170	119	221	17	136

Image Representation



Pixel of an RGB image are formed from the corresponding pixel of the three component images

Image Preprocessing

- Image preprocessing is the process of manipulating raw image data into a usable and meaningful format.
- It allows to eliminate unwanted distortions and enhance specific qualities essential for computer vision applications.
- Preprocessing is a crucial first step to prepare image data before feeding it into machine learning models.

Common methods:

- Image Resizing
- Image Normalization
- Image Denoising
- Image smoothing
- Image Sharpening
- Histogram Equalization
- Contrast Enhancement
- Image Augmentation

Goal:
*Improve **data quality for further analysis***.

Feature Extraction

It is the process of identifying and representing important structures in an image.

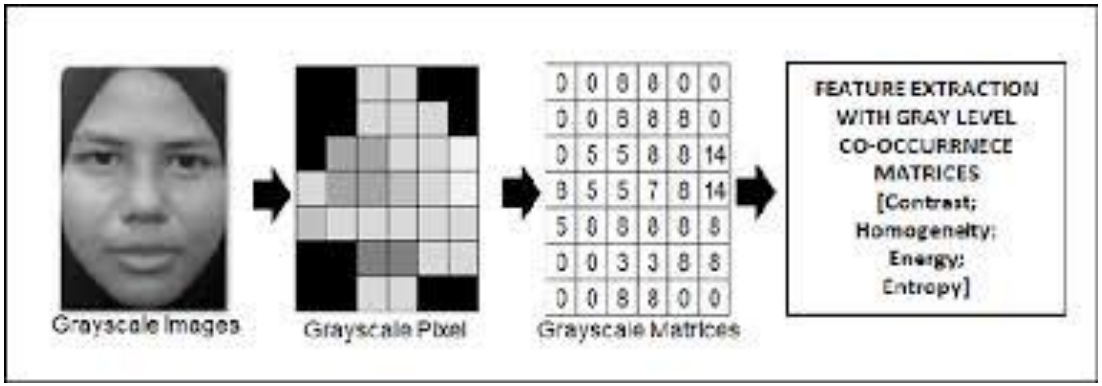
Instead of processing millions of raw pixels, computer vision systems *extract compact and meaningful representations of the image*.

Why Feature Extraction is Important

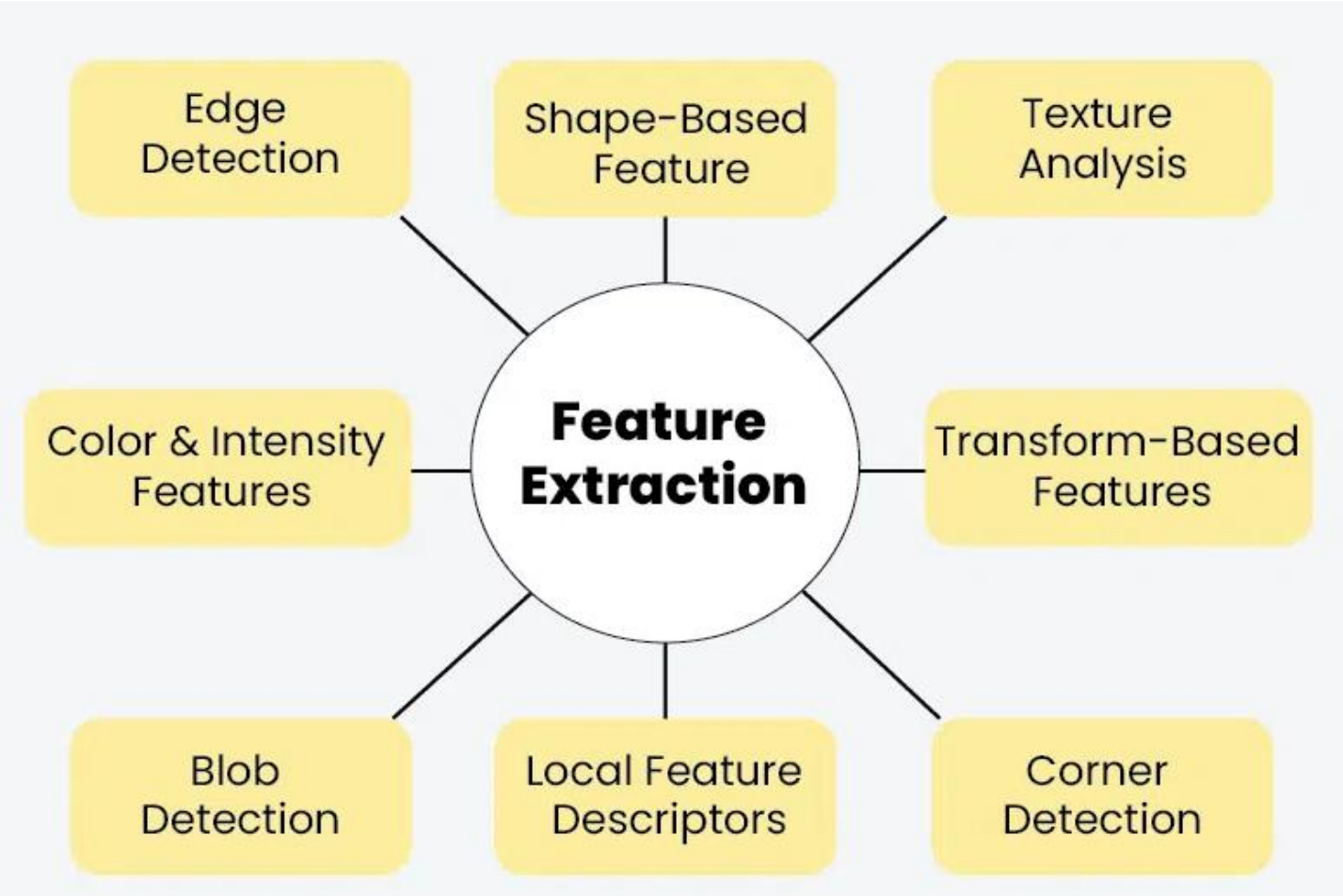
- ***Reduces dimensionality*** *of image data*
- *Improves* ***computational efficiency***
- *Helps algorithms* ***distinguish objects and patterns***

Applications:

- *Face recognition*
- *Medical imaging*
- *Object detection*
- *Autonomous vehicles*



Traditional (classical) Feature Extraction Techniques in Image Processing



Common Traditional Feature Extraction Techniques

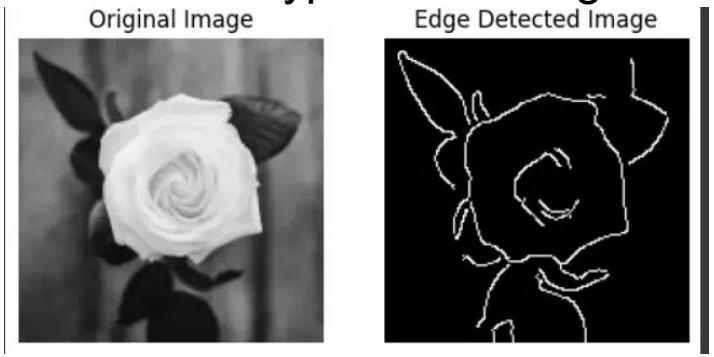
1. Edge & Corner Features: Detect structural boundaries and keypoints in images.

Examples

- Edge detection: Sobel, Canny
- Corner detection: Harris, Shi–Tomasi

Applications

Object boundaries, motion tracking



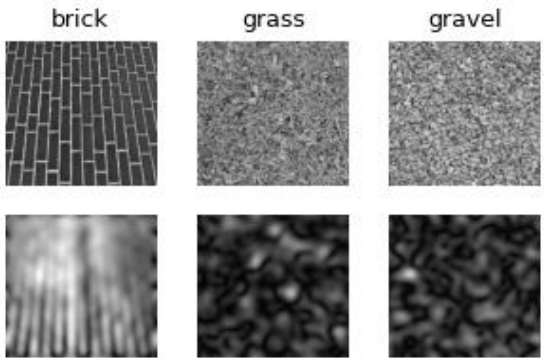
2. Texture Features: Capture repeating patterns and surface characteristics.

Examples

- Local Binary Pattern (LBP)
- Gabor filters

Applications

Material classification, defect detection



3. Shape Features : Describe geometric structures of objects.

Examples

- Contours
- Hough Transform
- Boundary descriptors

Applications

Object recognition, industrial inspection

Common Traditional Feature Extraction Techniques

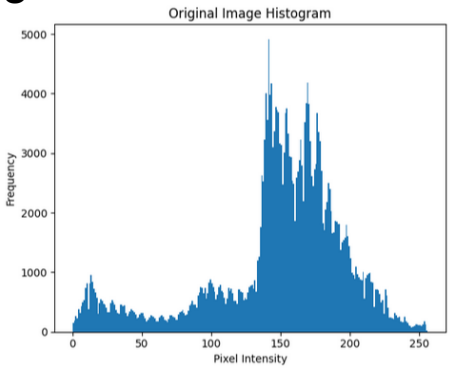
4. Color / Intensity Features: Analyze distribution of color and brightness.

Examples

- Color histograms
- Intensity histograms

Applications

Image retrieval, scene recognition



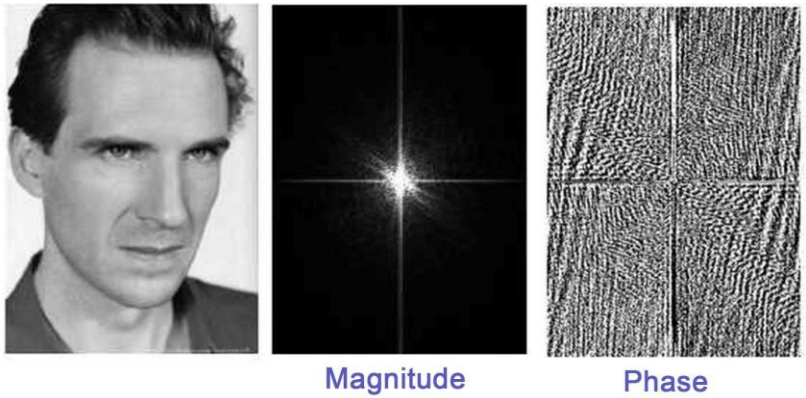
5. Transform-Based Features: Extract frequency information using mathematical transforms.

Examples

- Fourier Transform
- Wavelet Transform

Applications

Texture analysis, signal processing



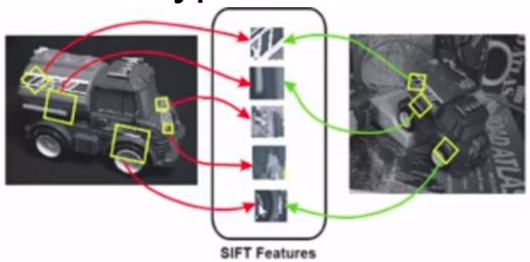
6. Local Feature Descriptors: Detect distinctive local keypoints for matching images.

Examples

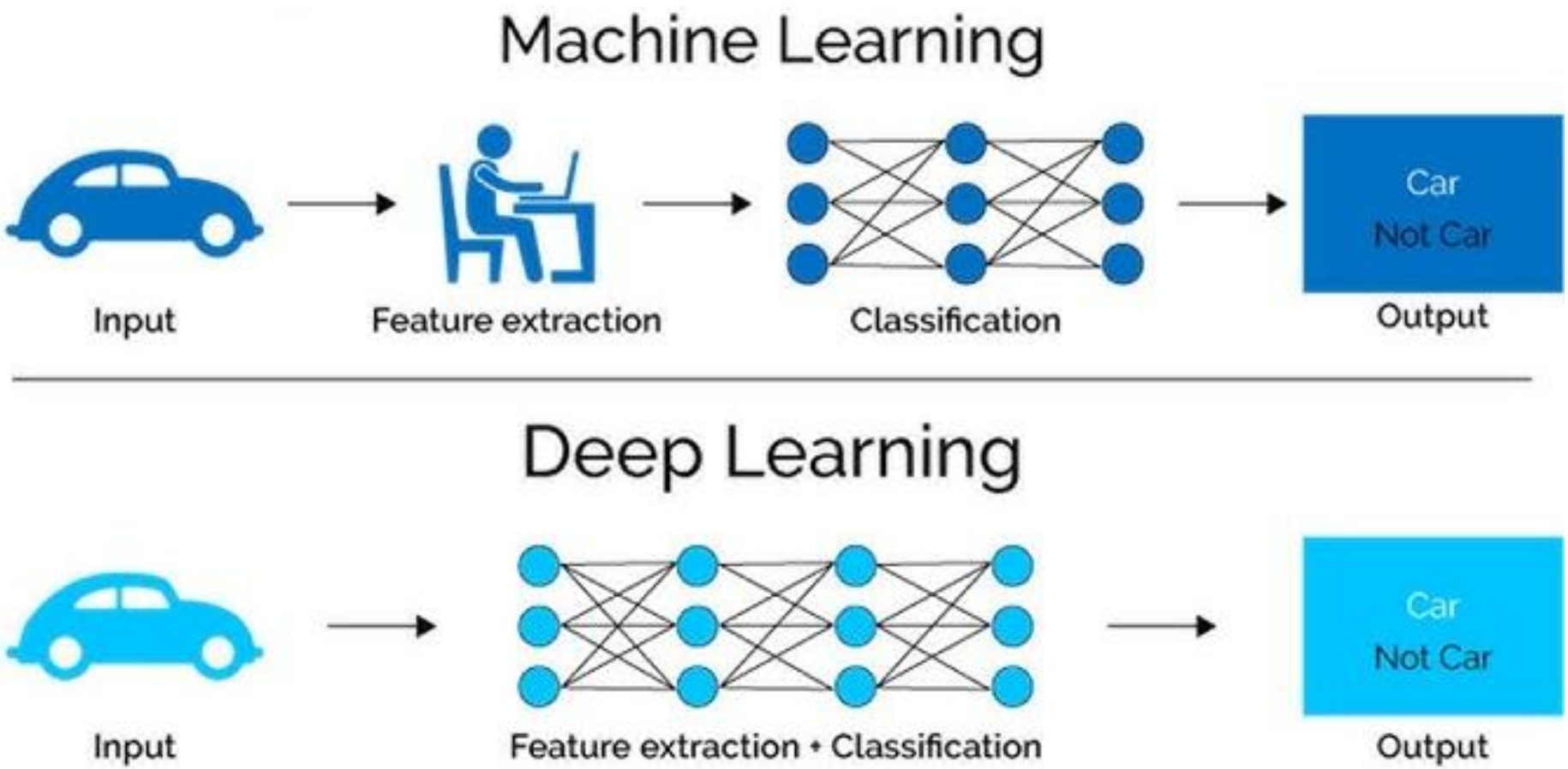
- SIFT

Applications

Image matching, panorama stitching, SLAM



Traditional Vs Deep Leaning Feature Extraction Techniques



Role of Pattern Recognition in Computer Vision

- Pattern recognition is the process of identifying patterns or objects in images using extracted features and algorithms. It perform tasks such as classification, detection, and segmentation.

Common pattern recognition tasks:

1. Image Classification

Assigns a label to the entire image

2. Object Detection

Identifies what objects are present and where they are located.

3. Image Segmentation

Divides the image into pixel-level regions representing different objects.

Pattern Recognition in Computer Vision

Image Classification

Traditional methods:

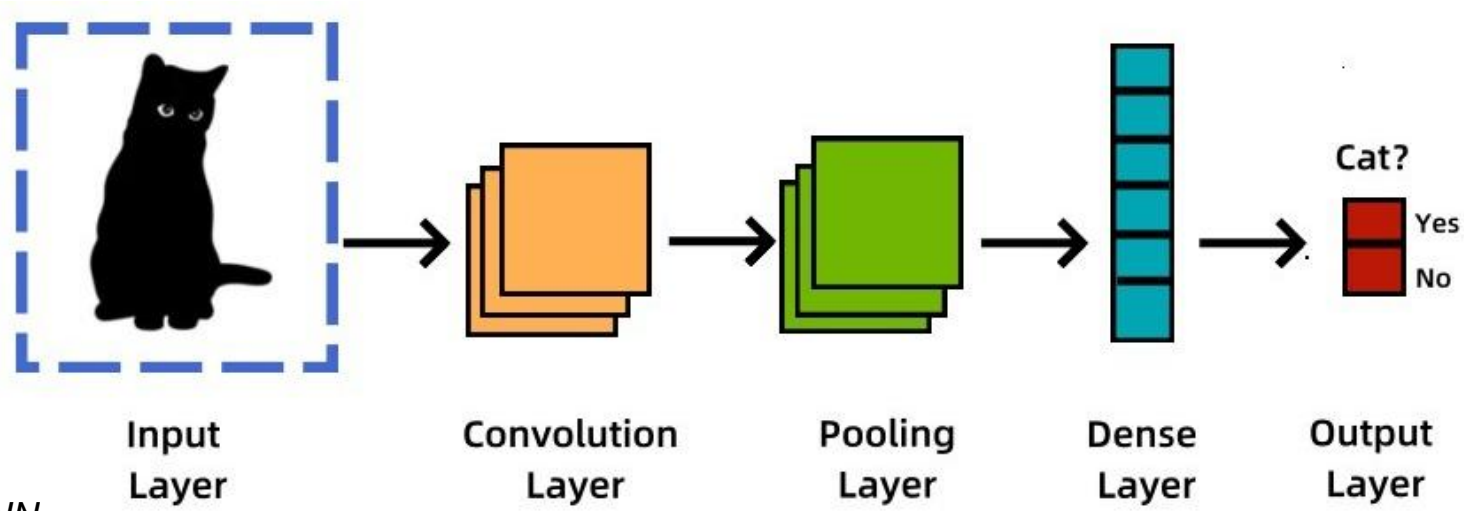
- SVM
- k-Nearest Neighbor
- Decision Trees

Deep learning methods:

- Convolutional Neural Networks (CNN)

Output

Class label for the image.



CNN

Pattern Recognition in Computer Vision

Object Detection

Traditional methods:

- Haar Cascades
- HOG + SVM

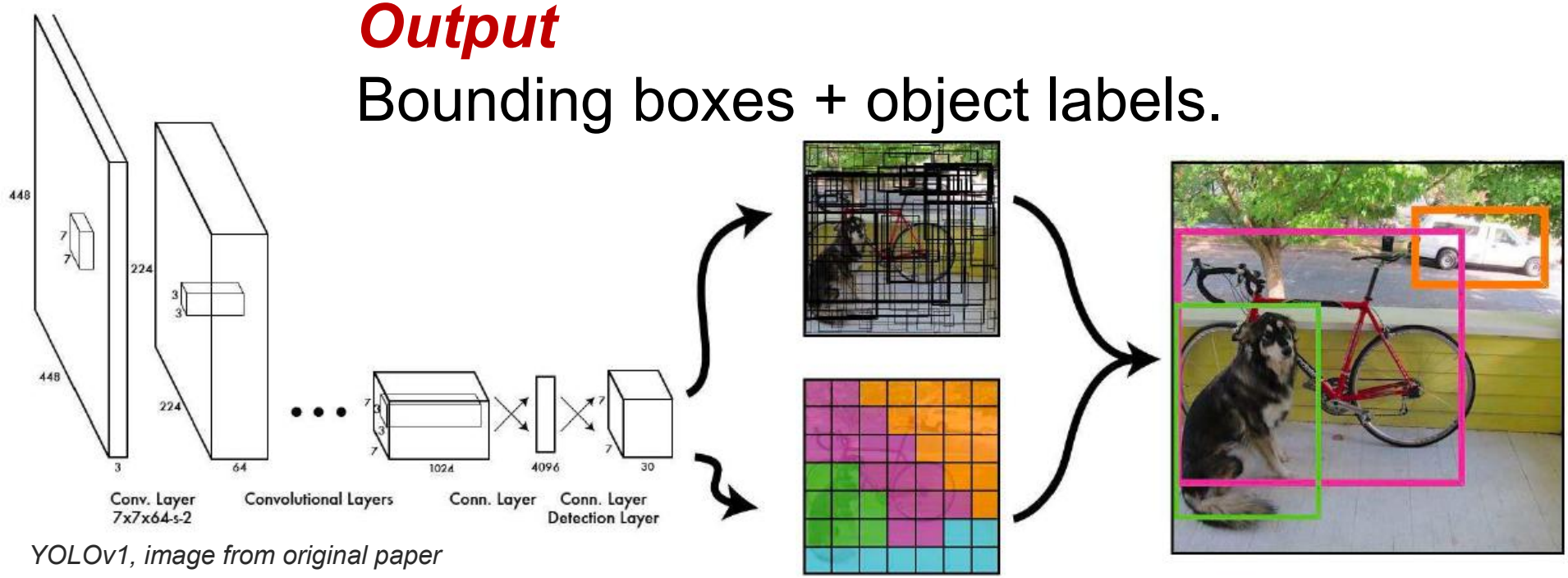
Histogram of Oriented Gradients (HOG)

Deep learning methods:

- YOLO (you look only once)
- Faster R-CNN
- SSD

Output

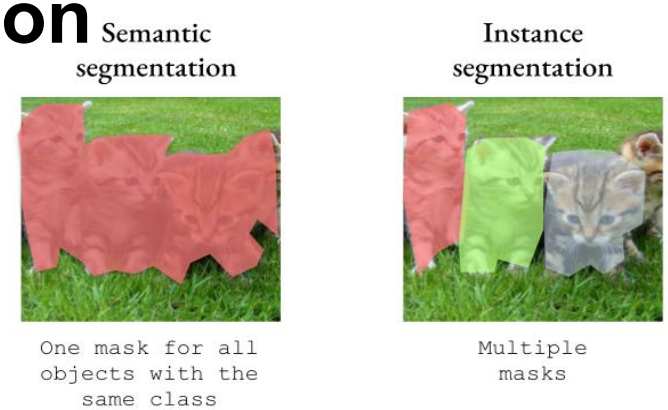
Bounding boxes + object labels.



YOLOv1, image from original paper

Pattern Recognition in Computer Vision

Image Segmentation

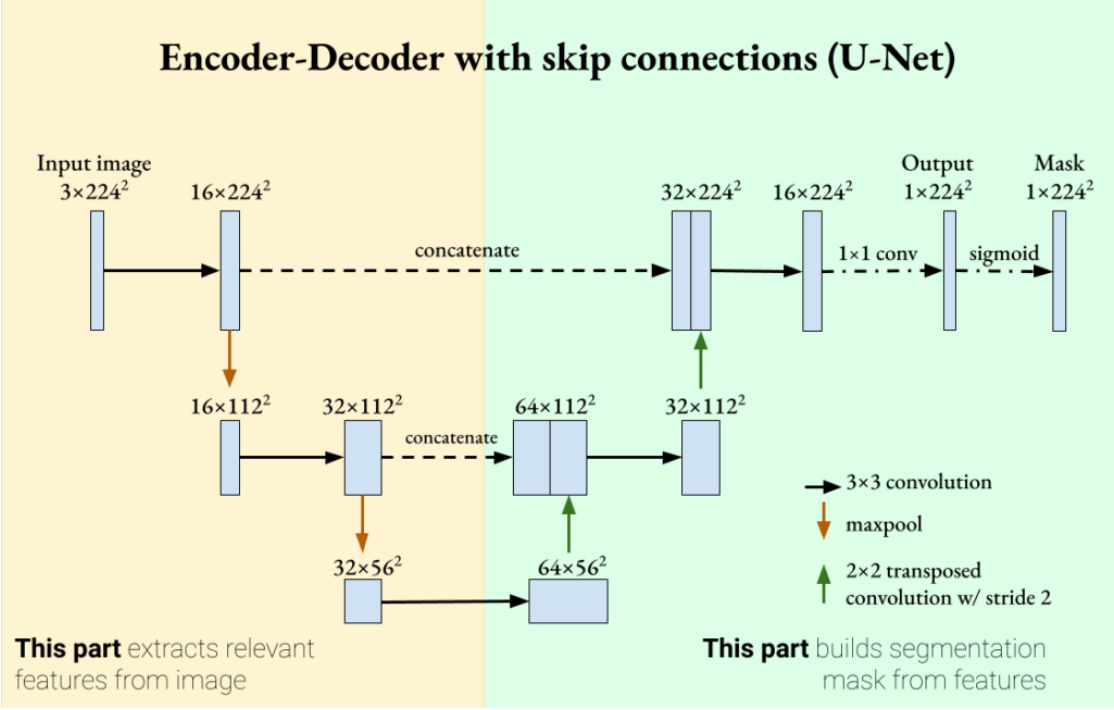


Traditional methods:

- Thresholding
- Edge-based segmentation
- Region growing
- Clustering (K-means)

Deep learning methods:

- U-Net
- Mask R-CNN



Output

Pixel-level labeled regions.